

Artha Drishti AI Agent

Detailed Architecture & Capability Report

1. Introduction

The Artha Drishti AI Agent is a professional Stock Market Assistant designed for the Indian financial markets (NSE/BSE). Built on top of LangChain and LangGraph, the agent harnesses the power of Google's Gemini LLMs to provide real-time market data, run trading strategies, analyze sentiment, and deliver AI-driven price predictions.

2. Core Architecture (LangGraph)

The agent is implemented as a state machine using LangGraph. This design enables deterministic cycles where the LLM decides whether to invoke tools, and loops back upon tool completion. State persistence is managed via PostgreSQL (using PostgresSaver), ensuring conversation history across sessions.

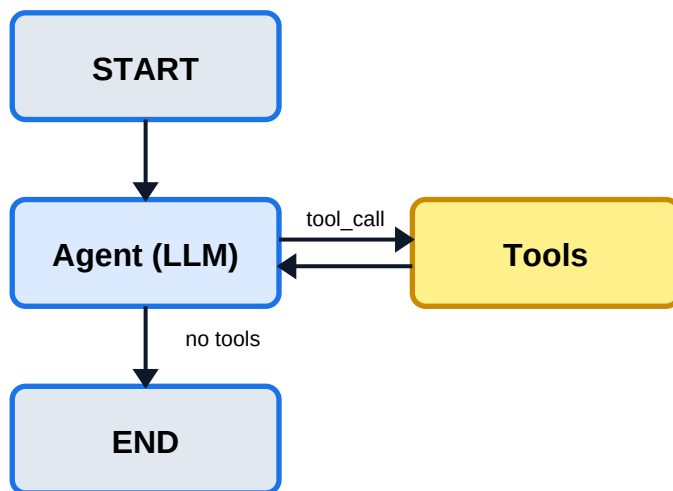


Figure 1: LangGraph State Machine Architecture of the AI Agent

3. Available Tools & Capabilities

To prevent hallucination, the agent strictly relies on specialized tools rather than its parametric memory for financial data. Below is the list of integrated tools:

Tool Name	Description	Primary Use Case
screen_stocks	Runs quantitative screening strategies (Piotroski, Momentum, Swing, Breakout, Value).	Finding stocks that match specific technical or fundamental criteria.
predict_stock	Fetches AI price predictions and technical analysis signals for a specific NSE ticker.	Forecasting price movements and assessing buy/sell signals.
get_market_overview	Retrieves the latest market snapshot with price and volume data for top stocks.	Providing a quick summary of current market conditions.
get_stock_history	Gets recent OHLCV (Open, High, Low, Close, Volume) history for the last 10 days.	Analyzing recent price action for a specific company.
get_sentiment	Analyzes news sentiment, calculating a score (-1 to 1) based on recent articles.	Gauging market mood and evaluating news impact on a stock.
generate_pattern_chart	Generates interactive Plotly candlestick charts with detected patterns.	Visualizing stock data and technical patterns directly in the chat UI.
search_web	Performs live web searches using Google Search API for grounded answers.	Fetching breaking news, IPO dates, or live prices not covered by other tools.

4. Prompt Engineering & Guardrails

The agent relies on a strict System Prompt with rules designed to ensure safety, accuracy, and professional conduct:

- **Anti-Hallucination:** The agent is instructed NEVER to fabricate prices, percentages, or dates. All numerical data must stem from a tool result.
- **Honesty:** If data is missing or a tool fails, the agent honestly admits its inability to fetch the information.
- **Mandatory Disclaimer:** The agent automatically appends a financial disclaimer to its responses: 'This is AI-generated analysis... not financial advice.'
- **UI Integration:** Special markdown tags (e.g., chart rendering tags) returned by tools are passed exactly as-is to ensure proper frontend rendering.

5. Data Flow and Persistence

The system employs a **streaming architecture**. As the LangGraph agent executes, chunks of text and tool invocation statuses are streamed immediately to the user via Server-Sent Events (SSE). The **PostgresSaver** checkpointing automatically saves the `MessagesState` to a PostgreSQL

database at the end of every execution node. This allows a user to refresh the page and continue their session seamlessly while capping the max history at a configurable limit (e.g., 20 messages) to save context window tokens.